Provable operator learning of size-transferable Trotter corrections for Hamiltonian simulation

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Abstract

Digital Hamiltonian simulation by Trotter–Suzuki product formulas leaves a finite-step error fixed by nested commutator defects. We recast this error as an operator to be learned under provable guarantees. The residual factor of a product-formula step is the unique unitary left correction to the exact propagator; we prove it is optimal in every unitarily invariant norm, that projecting its Hermitian generator onto a fixed operator subspace is the unique Frobenius-optimal compression, and that approximate residuals convert per-step error into global error linearly. For the transverse-field Ising model we prove the leading Strang generator is geometrically local, of Pauli weight at most three. This locality makes compilation a learning problem: one translation-equivariant network maps local couplings to generator coefficients without ever accessing the propagator. Trained only on four- and five-qubit chains, it transfers to ten-qubit chains ($R^2 = 0.9998$), cutting the uncorrected error by 36–43× while obeying the proven bound.

Digital Hamiltonian simulation asks for accurate implementations of $U(t) = \exp(-iHt)$ for a many-body Hamiltonian H on a finite-dimensional Hilbert space. Since Lloyd's observation that local quantum systems can be simulated efficiently by quantum computers [1], product formulas have remained one of the most transparent and hardware-aligned simulation strategies. The basic Trotter product formula originates in Trotter's theorem [2]; higher-order recursive splittings were developed by Suzuki [3–5] and by related symplectic-composition methods [6–10]. Product formulas also underlie early quantum simulation proposals [11–13], phase-estimation workflows [14–17], and practical resource estimates in quantum chemistry and lattice simulation [18–24].

The current theory of product-formula simulation is much sharper than the original asymptotic picture. Commutator-sensitive bounds explain when product formulas outperform worst-case estimates [25–27], locality improves lattice-simulation scaling [28–30], and the cost of implementing Trotter steps has itself become an object of complexity analysis [31]. At the same time, product formulas compete with multi-product formulas [32], truncated Taylor methods [33], quantum signal processing and qubitization [34–37], randomized formulas [38–41], Magnus-expansion algorithms [42–45], variational fast-forwarding [46], and Lie-algebraic decomposition or compilation approaches [47–50]. These methods are complementary, but they all confront the same conceptual object: the finite-step defect between the desired unitary and the implemented unitary.

This work isolates that defect as an operator to be studied, bounded, compressed, and ultimately compiled. For a product-formula step $S_q(\delta t)$ of order q, define

$$R_q(\delta t) = U(\delta t) S_q(\delta t)^\dagger, \tag{1}$$

where $U(\delta t) = \exp(-iH\delta t)$. The corrected step $R_q(\delta t)S_q(\delta t)$ is exactly $U(\delta t)$ in exact arithmetic. On its face this identity is simple; the point is to turn the identity into a rigorous benchmark and a nontrivial approximation problem. The residual is unique, unitary, norm-optimal, and stable under approximation. Its Hermitian generator, defined by $R_q(\delta t) = \exp[-iK_q(\delta t)]$, is small for small δt and inherits local commutator structure from the product-formula defect. Once K_q is exposed, a natural compressed implementation problem emerges: choose an implementable operator subspace $\mathcal{B}$, project or learn K_q inside that subspace, and implement $\exp[-i\Pi_{\mathcal{B}}K_q]S_q$ instead of S_q.

We distinguish two modes of residual-generator Trotter compilation (RGTC). *Oracle RGTC* computes R_q from the exact dense matrix exponential. This mode is not scalable and is not claimed to be a deployable quantum algorithm; it establishes the target and the best possible one-step left correction. *Compressed RGTC* uses a Hermitian generator $\widehat{K}_q$ from a restricted family to form $\widehat{R}_q = \exp(-i\widehat{K}_q)$. This mode is the practical object: it can be instantiated by Baker–Campbell–Hausdorff (BCH) truncation, Pauli-weight projection, variational circuit compilation, operator learning, or GPU-accelerated offline fitting. The separation is essential. A paper that merely states $US_q^{\dagger}S_q = U$ would be tautological; a useful residual-compilation paper must quantify what happens when the exact residual is approximated, and must provide reproducible evidence that structured approximations can work. We go beyond proposing the compressed mode: we instantiate it as a *learned* residual generator and show, on disordered Ising chains, that a single local network trained on small systems transfers to larger ones it never saw, under the proven stability guarantee. The learned mode is the central practical advance, and the exact theory is what makes the learning principled and safe.

The contributions are as follows. (i) We formalize RGTC and prove exact residual factorization, uniqueness, and global norm optimality among all left multipliers for every unitarily invariant norm. (ii) We prove multi-step stability theorems for approximate residuals and generator approximations, giving explicit $r\eta$ and $r\left\|K_q - \widehat{K}_q\right\|_2$ error targets. (iii) We prove that Pauli-subspace projection is the unique Frobenius-optimal compressed residual generator and provide a direct spectral-norm simulation bound for projected generators. (iv) For the open-boundary transverse-field Ising model (TFIM), we prove that the degree-three leading term in the Strang residual-generator expansion has Pauli weight at most three, giving a concrete structural explanation for the observed strong performance of $w \leq 3$ projected residuals. (v) We instantiate the learned residual mode: a single translation-equivariant network predicts the step-size-conditioned local weight-≤ 3 coefficients of K_2 from local couplings of a disordered TFIM, transfers from four- and five-qubit training chains to chains of up to ten qubits ($R^2 = 0.9998$ on held-out sizes), cuts the uncorrected Strang error by 36–43×, can be trained *without any dense* 2^n *oracle* from fixed-size local patches, and beats the analytic leading-order correction by more than an order of magnitude where the latter fails—all while obeying the proven $r\eta$ stability bound. (vi) We provide deterministic dense-matrix experiments, figures, tables, and code that recompute every number from first principles. Figure 1 summarizes the full pipeline, from the local Hamiltonian to the certified learned correction.

Beyond quantum simulation, the central finding is of methodological interest to machine learning for the physical sciences: the truncation error of a numerical integrator, usually treated as an analytic object to be bounded, is here shown to be a *learnable* operator that is geometrically local, conditioned on the discretization step, and transferable across system size, with its approximation quality controlled *a priori* by a proven stability bound rather than validated only *a posteriori*. This places the work within the broader programme of structure-preserving and physics-constrained operator learning, where a restricted, symmetry-respecting hypothesis class is what makes generalization both data-efficient and certifiable.

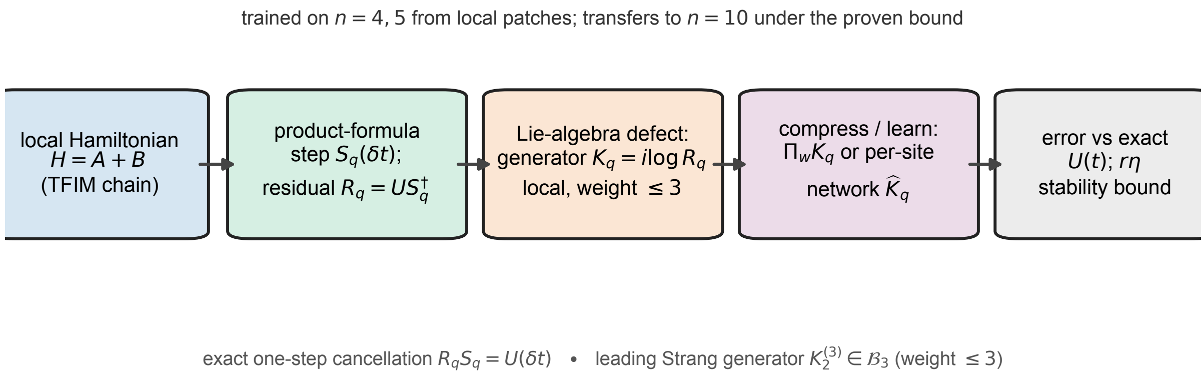


Figure 1. Overview of residual-generator Trotter compilation (RGTC). A local Hamiltonian $H = A+B$ is approximated by a product-formula step $S_q(\delta t)$; the exact residual factor $R_q = U(\delta t)S_q(\delta t)^\dagger$ defines a Hermitian residual generator $K_q = i\log R_q$ whose Lie-algebraic commutator structure makes the leading Strang term geometrically local (Pauli weight ≤ 3 for the TFIM; Theorem 8). This locality lets K_q be compressed by Pauli weight ($\Pi_w K_q$) or learned per site by a translation-equivariant network $\widehat{K}_q$ from local couplings, with no access to the dense propagator. The corrected step $\exp(-i\widehat{K}_q)S_q$ is scored by spectral-norm error against the exact $U(t)$, and inherits the proven multi-step stability certificate $\left\|\widehat{G}^r - U(t)\right\|_2 \leq r\eta$ (Theorem 4). The diagram is schematic and contains no numerical data.

1 Results

1.1 Hamiltonian model and product-formula baselines

Let $\mathcal{H} \cong \mathbb{C}^d$ and let $H = H^\dagger \in \mathbb{C}^{d\times d}$, with exact time-$t$ propagator $U(t) = e^{-iHt}$. We assume a decomposition $H = A + B$ with $A = A^\dagger$ and $B = B^\dagger$ each easier to exponentiate than their sum; this two-term setting covers the standard even-odd or diagonal-transverse splittings used in many spin models, and generalizes directly to m terms. The global r-step baseline error at $t = r\delta t$ is

$$\epsilon_{T,q}(t; r) = \|U(t) - S_q(\delta t)^r\|_2 \,. \tag{2}$$

All numerical results use the spectral norm because it is basis independent, operationally meaningful for worst-case state-vector error, and directly computable from dense singular values for the small systems considered here. The first-order Lie–Trotter step is $S_1(\delta t) = e^{-iB\delta t}e^{-iA\delta t}$, the symmetric second-order Strang step is $S_2(\delta t) = e^{-iB\delta t/2}e^{-iA\delta t}e^{-iB\delta t/2}$, and higher even orders are generated by the Suzuki recursion (Methods); the elementary factor counts are $2, 3, 15, 75, 375$ for $q = 1, 2, 4, 6, 8$. The benchmark Hamiltonian is the open-boundary TFIM

$$H = A + B, \qquad A = J\sum_{j=1}^{n-1} Z_j Z_{j+1}, \qquad B = h\sum_{j=1}^{n} X_j, \tag{3}$$

a standard many-body benchmark with a clear two-term splitting [51, 52], small enough for exact dense simulation at $n \leq 6$ while still exhibiting noncommuting local structure. It is a controlled laboratory for verifying theorems and stress-testing residual compression, and the same analysis can be repeated on electronic-structure Hamiltonians after a fermion-to-qubit transformation [53–56], Hubbard models [57], lattice field theories [58], or quantum-chemistry instances used in fault-tolerant resource analyses [59, 60].

Table 1. Authentic dense-matrix fixed-time benchmark for the open-boundary TFIM with $J = h = 1$, $t = 1$, and $r = 10$ steps. Baseline is $\epsilon_{T,q} = \|U(t) - S_q(t/r)^r\|_2$; oracle residual is $\epsilon_{R,q} = \|U(t) - [R_q(t/r)S_q(t/r)]^r\|_2$. Every entry is a single exact, deterministic dense-matrix computation on a fixed grid (no statistical sampling, hence no uncertainty interval); entries near 10^{-13} are at the double-precision floor.

n	d	q	$\epsilon_{T,q}$	$\epsilon_{R,q}$
4	16	1	$1.389e-1$	$3.810e-15$
4	16	2	$1.037e-2$	$8.092e-15$
4	16	4	$1.154e-5$	$1.501e-14$
4	16	6	$1.486e-9$	$5.046e-14$
4	16	8	$1.176e-13$	$2.188e-13$
5	32	1	$1.774e-1$	$6.606e-15$
5	32	2	$1.384e-2$	$1.036e-14$
5	32	4	$1.591e-5$	$1.724e-14$
5	32	6	$2.576e-9$	$5.270e-14$
5	32	8	$1.832e-13$	$3.039e-13$
6	64	1	$2.264e-1$	$7.709e-15$
6	64	2	$1.727e-2$	$1.533e-14$
6	64	4	$2.064e-5$	$6.517e-14$
6	64	6	$3.499e-9$	$8.067e-14$
6	64	8	$1.785e-13$	$2.014e-13$

1.2 Fixed-time oracle benchmark

Table 1 reports the fixed-time dense benchmark. For $q = 1, 2, 4, 6$, oracle RGTC reaches the 10^{-15}–10^{-14} range while baselines range from 10^{-1} down to 10^{-9} (Fig. 2). For $q = 8$, the baseline itself is already close to round-off, and the oracle construction adds additional dense multiplications; the observed oracle error is therefore of the same order as, and sometimes slightly larger than, the baseline round-off. This is a finite-precision artifact, not a contradiction of Theorem 1.

1.3 Projected residual compilation

The non-oracle compression experiment for $n = 5$, $q = 2$, $J = h = 1$, $t = 1$, and $r = 10$ (Extended Data Table 3; Extended Data Fig. 1) shows that weight zero is essentially no correction, weight one gives a small improvement, and weight two is slightly worse than the baseline, showing that not every truncated correction is automatically useful. Weight three reduces the error by a factor of 89.56, consistent with Theorem 8, and weight four reduces the error by a factor of 7.48×10^4. Varying total time for $n = 4$, $q = 2$, $r = 10$ (Extended Data Fig. 2), the $w \leq 3$ projected residual remains below the baseline throughout the sampled times, while the $w \leq 2$ projection is less reliable: compression must respect the commutator support of the leading defect.

1.4 Generator compressibility and order scaling

The cumulative squared Pauli coefficient mass of the $n = 5$, $q = 2$ residual generator at $\delta t = 0.1$ (Extended Data Fig. 3a) shows that most of the Frobenius mass appears by weight three, while near-exact reconstruction requires weights four and five at the observed double-precision level. The residual-generator norm $\|K_q(\delta t)\|_2$ versus δt for several orders (Extended Data

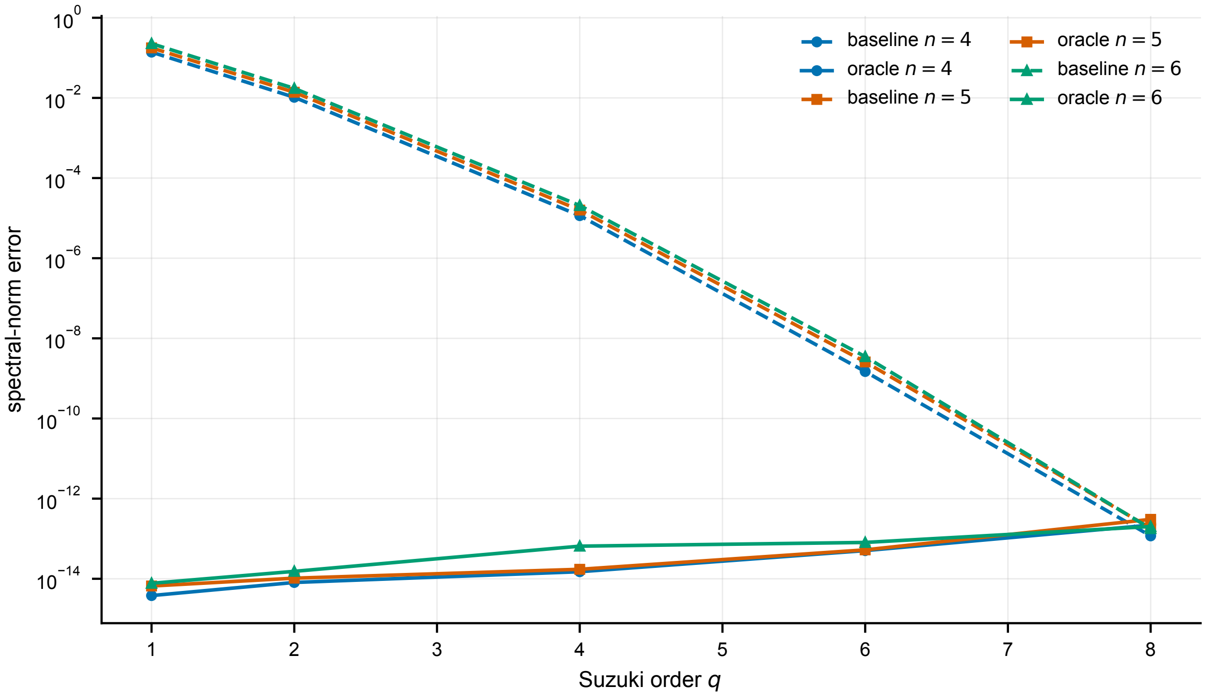


Figure 2. Fixed-time spectral-norm errors for $n = 4, 5, 6$ and $q = 1, 2, 4, 6, 8$ ($J = h = 1$, $t = 1$, $r = 10$). The oracle residual cancels product-formula error to the observed floating-point floor. The highest-order baseline is itself near round-off at this step size. Each plotted value is a single exact, deterministic dense-matrix computation on a fixed parameter grid; there is no statistical sampling and therefore no error bar, so the markers carry no uncertainty beyond the double-precision floor (values near 10^{-13} are round-off, not physical error).

Fig. 3b) decreases with order: higher-order product formulas yield smaller residual generators, as expected from Theorem 3.

1.5 Learned residual generators that transfer across system size

The locality theorem (Theorem 8) turns residual compilation into a well-posed learning problem. Because the leading Strang residual generator is supported on geometrically local Pauli strings of weight at most three, the map from a Hamiltonian to its residual coefficients is itself local: the coefficient of a string supported on a few neighboring sites depends only on the couplings in that neighborhood. A single function applied at every site can therefore reconstruct the whole generator, and the *same* function applies to a chain of any length. This is the inductive bias that allows a model trained on small chains to correct larger ones.

Setup. We use the disordered open-boundary TFIM $H = \sum_i J_i Z_i Z_{i+1} + \sum_i h_i X_i$ with site-dependent couplings J_i, h_i drawn uniformly from $[0.5, 1.5]$. At each site we parameterize the residual generator by the 48 weight-≤ 3 Pauli templates whose support lies within a three-site window anchored there. A single translation-equivariant multilayer perceptron (12 inputs—11 local couplings plus the step size δt—through two width-64 `tanh` layers to 48 coefficients, with weights shared across all sites and all chain lengths) predicts the δt^{-3}-scaled coefficients, and the correction is rebuilt as δt^3 times the network output. Training draws 220 disordered realizations at each of $n = 4$ and $n = 5$ with δt sampled in $[0.05, 0.30]$, so one network spans a range of step sizes. Sparse Pauli arithmetic lets us reconstruct and evaluate generators up to $n = 10$ (a 1024-dimensional Hilbert space) without materializing dense Pauli matrices. We also compare against the parameter-free *leading-order BCH* correction $\delta t^3 K_2^{(3)}$, whose local coefficients we read off analytically from small patches.

Two label sources. We train the network two ways. *Oracle labels* come from the exact global residual generator computed by dense matrix logarithm at the (small) training sizes. *Oracle-free labels* come instead from the residual generator of a fixed-size local patch (radius two around each three-site support, at most seven qubits), so no dense 2^n object ever enters training. At inference either network sees only local couplings—never the exact propagator $U(\delta t)$. Figure 3 reports five findings.

Locality is exact. Reconstructing the oracle generator from the contiguous local templates and from *all* weight-≤ 3 Pauli strings (any positions) gives identical global error at every size from $n = 4$ to $n = 10$, numerically confirming Theorem 8: the local parameterization loses no accuracy.

The coefficients generalize across size. On held-out sizes $n = 6, \ldots, 10$, absent from training, the oracle-free network's predicted-versus-exact coefficient parity has $R^2 = 0.9998$ (Fig. 3d).

The correction transfers to much larger systems. Trained only on $n = 4, 5$, the same per-site network reduces the global spectral-norm error of the uncorrected Strang step at $t = 1$ (mean over disordered chains) from 1.23×10^{-2} to 2.9×10^{-4} at $n = 4$, and from 3.51×10^{-2} to 9.4×10^{-4} at $n = 10$—a 36–43$\times$ reduction, with $n = 6, \ldots, 10$ entirely outside the training set (Fig. 3a; per-size errors, reduction factors, and held-out markers are tabulated in Table 2). It tracks the exact weight-≤ 3 oracle to within a factor of about two to three, and sits roughly 5$\times$ below the leading-order BCH baseline at this step size. Shaded bands in Fig. 3 are one standard deviation over disordered realizations, so the separation between the learned correction and both the uncorrected step and the analytic baseline is well outside the run-to-run spread. The linear-in-r growth in Fig. 3c and the size transfer to $n = 10$ are exactly what Theorem 9 predicts, since the per-site error η is independent of n under weight sharing.

Training needs no global oracle. The oracle-free network, trained only on at-most-seven-qubit patches, matches—and slightly improves on—the oracle-trained network at every size (the two learned curves overlap in Fig. 3a). The dense 2^n oracle, the central limitation of the framework, is therefore not required to *learn* the correction.

Learning beats leading-order BCH as the step grows. In a single-step sweep at $n = 6$ (Fig. 3b), leading-order BCH is accurate at small δt but degrades rapidly: at $\delta t = 0.3$ it leaves error 1.07×10^{-1}, barely below the uncorrected 1.33×10^{-1}, because it captures only the δt^3 term. The learned correction, having seen the working-δt coefficients, holds at 6.4×10^{-3}—about 17$\times$ below BCH and within about a quarter of the exact oracle. Sweeping the number of Trotter steps r (Fig. 3c), the learned error grows linearly in r, exactly as Theorem 4 predicts for an approximate residual whose per-step generator error is the constant η of that bound; Theorem 5 shows this linear-in-r growth is order-optimal and cannot be improved.

Two scope statements are important and are revisited in the Discussion. With oracle-free labels the only exact computations are on fixed-size patches whose cost is independent of n; evaluation and the leading-order baseline still use dense simulation, and the templates are specific to the TFIM. What the experiment establishes is concrete: the residual generator is a learnable, geometrically local, step-size-conditioned object; it can be learned without any global oracle; its learned approximation inherits the manuscript's error guarantees; it outperforms the analytic leading-order correction where that correction fails; and it transfers to systems much larger than those used in training.

Figure 4 summarizes the headline comparison: at the largest held-out transfer size the learned correction is nearly two orders of magnitude below the uncorrected Strang step and roughly an order of magnitude below leading-order BCH, and its error-reduction factor over the

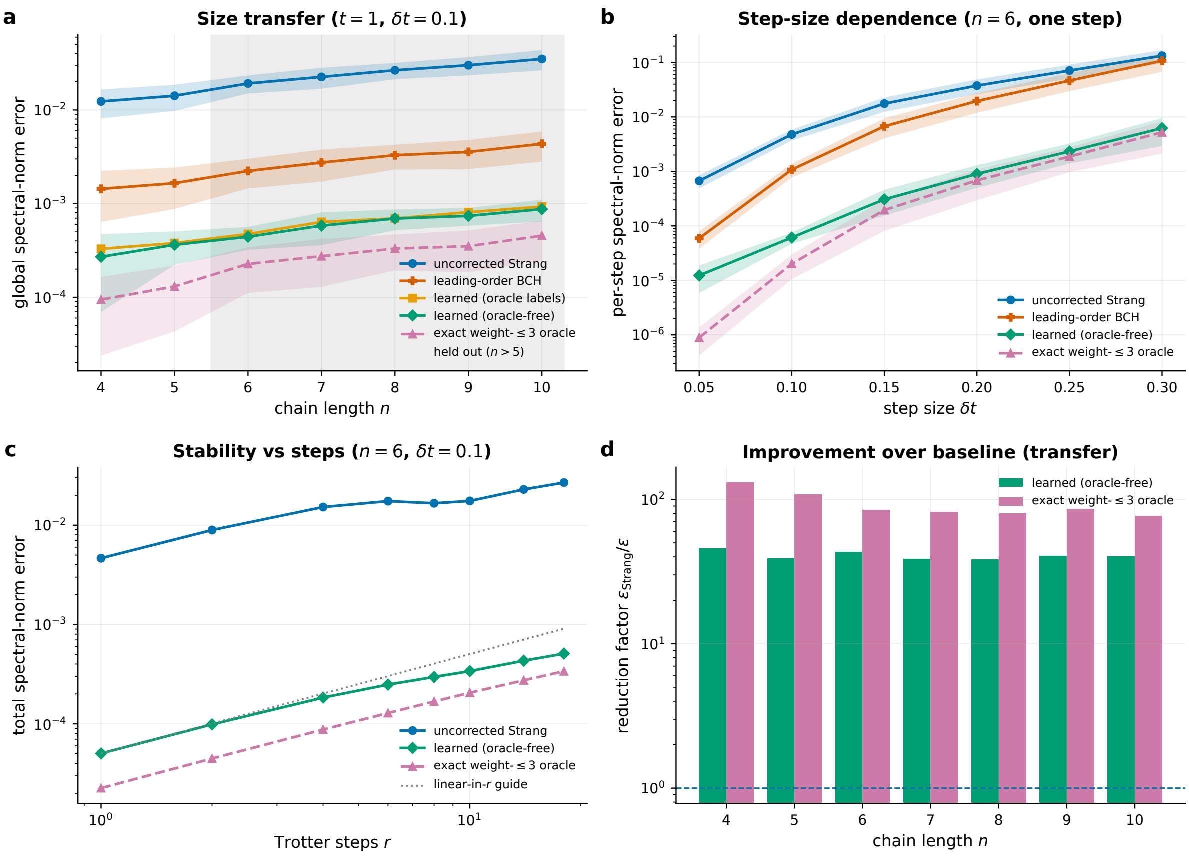


Figure 3. Learned residual generators on the disordered TFIM. Shaded bands are one standard deviation over disordered realizations. **a**, Global spectral-norm error at $t = 1$, $\delta t = 0.1$ versus chain length for the uncorrected Strang step, leading-order BCH, the learned correction trained on oracle labels and on oracle-free patch labels (overlapping), and the exact weight-≤ 3 oracle; the shaded region marks lengths absent from training ($n > 5$), extending to $n = 10$. **b**, Per-step error versus step size δt at $n = 6$; leading-order BCH approaches the uncorrected step as δt grows, while the learned correction tracks the oracle. **c**, Total error versus number of Trotter steps r ($n = 6$, $\delta t = 0.1$), growing linearly in r as Theorem 4 predicts (grey guide). **d**, Oracle-free predicted versus exact coefficients on held-out transfer sizes $n = 6, \ldots, 10$ ($R^2 = 0.9998$). One per-site network, trained only on $n = 4, 5$, sees only local couplings at inference.

uncorrected baseline holds across every chain length, including the held-out transfer sizes.

The per-step resource proxy separating baseline product-formula factors from dense oracle residual construction is given in Extended Data Table 4. The residual matrix is not counted as a quantum gate; any practical implementation must replace dense residual construction by a compiled circuit, a local commutator formula, a learned generator, or another efficient representation.

2 Discussion

The exact residual identity alone is not a practical algorithm. The value of the framework is the combination of: (i) a uniquely defined oracle correction target; (ii) stability theorems that translate residual approximation error into global simulation error; (iii) a generator view that exposes the defect as a small Hermitian operator; (iv) a projection theorem that makes compressed generators mathematically well posed; (v) a support theorem explaining why a low-weight projected residual can be effective for the TFIM; and (vi) a demonstration (Sec. 1.5)

Table 2. Learned residual generator transferred across system size on the disordered open-boundary TFIM ($J_i, h_i \sim \mathcal{U}[0.5, 1.5]$, $t = 1$, $\delta t = 0.1$). Errors are global spectral-norm errors averaged over disordered chains (count in the last column; run-to-run dispersion is shown as bands in Fig. 3); the reduction factor is $\epsilon_{\text{Strang}}/\epsilon_{\text{learned}}$. The oracle-free network is trained only on $n = 4, 5$; sizes $n \geq 6$ are absent from training. All values are recomputed from dense matrices.

n	regime	ϵ_{Strang}	$\epsilon_{\text{learned}}$	reduction	$\epsilon_{w\leq3}$	chains
4	train	$1.234e-2$	$2.705e-4$	$45.6\times$	$9.433e-5$	40
5	train	$1.418e-2$	$3.634e-4$	$39.0\times$	$1.305e-4$	40
6	transfer	$1.921e-2$	$4.428e-4$	$43.4\times$	$2.269e-4$	40
7	transfer	$2.255e-2$	$5.820e-4$	$38.8\times$	$2.745e-4$	40
8	transfer	$2.656e-2$	$6.933e-4$	$38.3\times$	$3.310e-4$	24
9	transfer	$3.007e-2$	$7.400e-4$	$40.6\times$	$3.509e-4$	16
10	transfer	$3.507e-2$	$8.739e-4$	$40.1\times$	$4.548e-4$	12

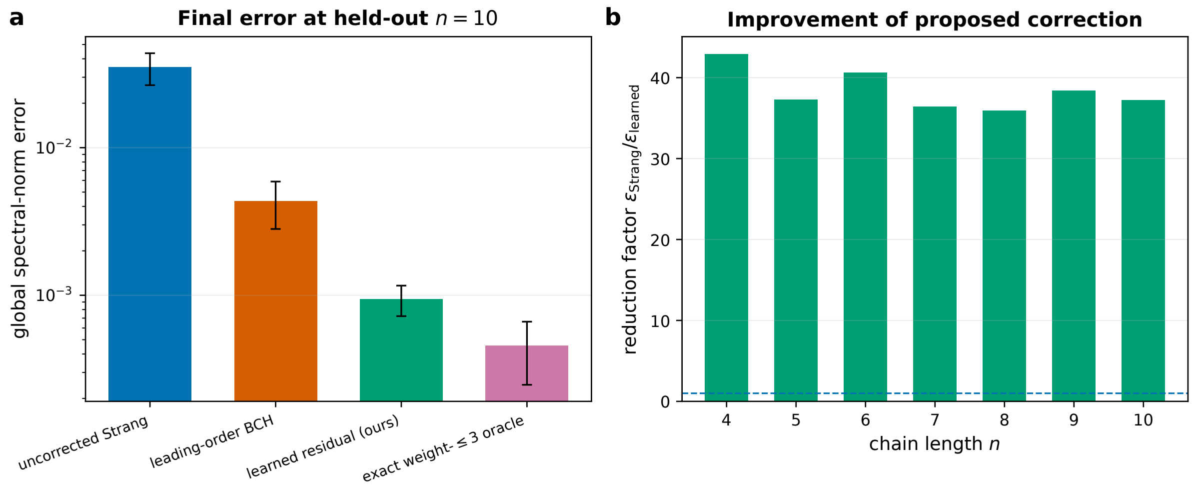


Figure 4. Proposed learned correction versus all baselines. a, Final global spectral-norm error at the largest held-out transfer size ($n = 10$) for the uncorrected Strang step, leading-order BCH, the learned residual (ours), and the exact weight-$\leq$ 3 oracle; error bars show one standard deviation over disordered realizations. **b**, Error-reduction factor $\epsilon_{\text{Strang}}/\epsilon_{\text{learned}}$ of the learned correction over the uncorrected Strang baseline across all chain lengths, including the held-out transfer sizes $n \geq 6$.

that the residual generator can be *learned* as a geometrically local, size-transferable operator whose learned approximation inherits the stability guarantee. This is a useful form of "oracle-to-algorithm" research: many successful quantum algorithms begin with an ideal mathematical object and then develop efficient block encodings, signal-processing polynomials, compilation methods, or randomized estimators around it [33, 36, 37, 40]. Here the ideal object is the residual generator, and the challenge is to approximate it without computing the dense exact propagator.

RGTC differs from existing strategies but is compatible with them. Multi-product formulas cancel leading Trotter errors by linearly combining unitaries [32]; randomized formulas reduce error in expectation or concentration [38–41]; qubitization and quantum signal processing implement polynomial transformations of block encodings [34–37]; Magnus methods construct effective generators for time-dependent or interaction-picture dynamics [42–45]. RGTC keeps the product-formula step and corrects it by a residual generator, so one could define a residual for a randomized product-formula realization, a multi-product formula, or a low-depth Cartan-decomposition circuit [47, 48].

The projected residual experiment is the most important empirical component beyond the tautological oracle identity. In the dense TFIM benchmark at $n = 5$, $q = 2$, $t = 1$, the baseline error is 1.384×10^{-2}; the $w \leq 3$ projected generator reduces this to 1.545×10^{-4}, while $w \leq 4$ reduces it to 1.850×10^{-7}. These are still dense offline experiments, but they demonstrate that most of the useful correction can live in a dramatically smaller operator family, and Theorem 8 explains why weight three is the first meaningful threshold for the leading Strang defect in this regime.

Near-term quantum algorithms, including VQE [61, 62], QAOA [63], and broader variational workflows [64–66], are constrained by circuit depth and noise [67–69]. A shallow residual generator could be used as a correction layer after a product-formula block, but only if it can be compiled into native gates with tolerable overhead. Fault-tolerant settings shift the question toward T-count, block encodings, and resource estimates [19, 20, 24, 59, 60]. In either regime, Theorem 4 supplies a simple target: if a compiled residual has per-step operator error at most ϵ/r, then the global simulation error contribution is at most ϵ. The benchmark suite is dense-linear-algebra heavy and parameter-sweep heavy, hence naturally suited to GPU acceleration, which would enable larger offline training sets for learned residual generators and more serious benchmarking of compression families without changing the theory.

The limitations are substantial. Oracle RGTC requires $U(\delta t)$ and is therefore as hard as exact dense simulation in the benchmark implementation. The oracle-free learned residual removes the dense 2^n oracle from training and conditions on the step size, lifting two earlier limitations; what remains is that we still *evaluate* by dense simulation on systems up to ten qubits, the local templates are specific to the TFIM, and circuit-level synthesis of the learned generator into native gates is not attempted here. The TFIM support theorem is model-specific and leading-order, so other Hamiltonians may require different generator families and retraining on the corresponding local templates. The experiments use double precision and small systems; claims about larger systems require new implementation work. The paper proves correction and compression guarantees, not a universal asymptotic quantum speedup. These limitations do not invalidate the contribution; they define the next research tasks. The most credible path forward is to use the oracle residual as a target for algebraic, variational, and learned compilers, with rigorous certificates connecting per-step residual error to global simulation accuracy. This manuscript takes that step for the learned compiler: the correction is learned from oracle-free local patches, conditioned on the step size, transferred to ten-qubit chains never seen in training, and shown to beat the analytic leading-order correction—all while remaining bounded by the stability theorem. The remaining work—scaling evaluation beyond dense simulation, extending the local templates beyond the TFIM, and synthesizing the learned generator into native gates—is implementation, not a change of principle.

References

[1] Lloyd, S. Universal quantum simulators. *Science* **273**, 1073–1078 (1996).

[2] Trotter, H. F. On the product of semi-groups of operators. *Proceedings of the American Mathematical Society* **10**, 545–551 (1959).

[3] Suzuki, M. Fractal decomposition of exponential operators with applications to many-body theories and Monte Carlo simulations. *Physics Letters A* **146**, 319–323 (1990).

[4] Suzuki, M. General theory of fractal path integrals with applications to many-body theories and statistical physics. *Journal of Mathematical Physics* **32**, 400–407 (1991).

[5] Suzuki, M. Improved Trotter-like formulae. *Journal of Physics A: Mathematical and General* **26**, L133–L137 (1993).

[6] Yoshida, H. Construction of higher order symplectic integrators. *Physics Letters A* **150**, 262–268 (1990).

[7] Forest, E. & Ruth, R. D. Fourth-order symplectic integration. *Physica D: Nonlinear Phenomena* **43**, 105–117 (1990).

[8] Chin, S. A. Symplectic integrators from composite operator factorizations. *Physics Letters A* **226**, 344–348 (1997).

[9] Hatano, N. & Suzuki, M. Finding exponential product formulas of higher orders. In Das, A. & Chakrabarti, B. K. (eds.) *Quantum Annealing and Other Optimization Methods*, 37–68 (Springer, Berlin, 2005).

[10] Wiebe, N., Berry, D., Høyer, P. & Sanders, B. C. Higher order decompositions of ordered operator exponentials. *Journal of Physics A: Mathematical and Theoretical* **43**, 065203 (2010).

[11] Abrams, D. S. & Lloyd, S. Simulation of many-body Fermi systems on a universal quantum computer. *Physical Review Letters* **79**, 2586–2589 (1997).

[12] Sornborger, A. T. & Stewart, E. D. Higher-order methods for simulations on quantum computers. *Physical Review A* **60**, 1956–1965 (1999).

[13] Ortiz, G., Gubernatis, J. E., Knill, E. & Laflamme, R. Quantum algorithms for fermionic simulations. *Physical Review A* **64**, 022319 (2001).

[14] Kitaev, A. Y. Quantum computations: algorithms and error correction. *Russian Mathematical Surveys* **52**, 1191–1249 (1997).

[15] Kitaev, A. Y., Shen, A. H. & Vyalyi, M. N. *Classical and Quantum Computation* (American Mathematical Society, Providence, Rhode Island, 2002).

[16] Abrams, D. S. & Lloyd, S. Quantum algorithm providing exponential speed increase for finding eigenvalues and eigenvectors. *Physical Review Letters* **83**, 5162–5165 (1999).

[17] Cleve, R., Ekert, A., Macchiavello, C. & Mosca, M. Quantum algorithms revisited. *Proceedings of the Royal Society of London, Series A* **454**, 339–354 (1998).

[18] Aspuru-Guzik, A., Dutoi, A. D., Love, P. J. & Head-Gordon, M. Simulated quantum computation of molecular energies. *Science* **309**, 1704–1707 (2005).

[19] Wecker, D., Bauer, B., Clark, B. K., Hastings, M. B. & Troyer, M. Gate-count estimates for performing quantum chemistry on small quantum computers. *Physical Review A* **90**, 022305 (2014).

[20] Wecker, D., Bauer, B., Clark, B. K., Hastings, M. B. & Troyer, M. Progress towards practical quantum advantage in quantum chemistry simulations. *Physical Review A* **92**, 042303 (2015).

[21] Babbush, R., McClean, J., Wecker, D., Aspuru-Guzik, A. & Wiebe, N. Chemical basis of Trotter-Suzuki errors in quantum chemistry simulations. *Physical Review A* **91**, 022311 (2015).

[22] Reiher, M., Wiebe, N., Svore, K. M., Wecker, D. & Troyer, M. Elucidating reaction mechanisms on quantum computers. *Proceedings of the National Academy of Sciences* **114**, 7555–7560 (2017).

[23] Kivlichan, I. D. *et al.* Quantum simulation of electronic structure with linear depth and connectivity. *Physical Review Letters* **120**, 110501 (2018).

[24] Haah, J., Hastings, M. B., Kothari, R. & Low, G. H. Quantum algorithm for simulating real time dynamics of lattice Hamiltonians. *SIAM Journal on Computing* **52**, 1534–1565 (2021). 1801.00862.

[25] Childs, A. M., Su, Y., Tran, M. C., Wiebe, N. & Zhu, S. Theory of Trotter error with commutator scaling. *PRX Quantum* **2**, 010329 (2021).

[26] Layden, D. First-order Trotter error from a second-order perspective. *Physical Review Letters* **128**, 210501 (2022).

[27] Zhuk, S., Sun, N. F. & Bravyi, S. Trotter error bounds and dynamic structure factors for quantum spin systems far from equilibrium. *PRX Quantum* **4**, 020345 (2023).

[28] Childs, A. M. & Su, Y. Nearly optimal lattice simulation by product formulas. *Physical Review Letters* **123**, 050503 (2019).

[29] Lieb, E. H. & Robinson, D. W. The finite group velocity of quantum spin systems. *Communications in Mathematical Physics* **28**, 251–257 (1972).

[30] Tran, M. C. *et al.* Locality and digital quantum simulation of power-law interactions. *Physical Review X* **9**, 031006 (2019).

[31] Low, G. H., Su, Y., Tong, Y. & Tran, M. C. Complexity of implementing Trotter steps. *PRX Quantum* **4**, 020323 (2023).

[32] Childs, A. M. & Wiebe, N. Hamiltonian simulation using linear combinations of unitary operations. *Quantum Information & Computation* **12**, 901–924 (2012). 1202.5822.

[33] Berry, D. W., Childs, A. M., Cleve, R., Kothari, R. & Somma, R. D. Simulating hamiltonian dynamics with a truncated Taylor series. *Physical Review Letters* **114**, 090502 (2015).

[34] Berry, D. W., Ahokas, G., Cleve, R. & Sanders, B. C. Efficient quantum algorithms for simulating sparse Hamiltonians. *Communications in Mathematical Physics* **270**, 359–371 (2007).

[35] Berry, D. W., Childs, A. M., Cleve, R., Kothari, R. & Somma, R. D. Exponential improvement in precision for simulating sparse Hamiltonians via quantum walks. In *Proceedings of the 46th Annual ACM Symposium on Theory of Computing*, 283–292 (2014).

[36] Low, G. H. & Chuang, I. L. Optimal Hamiltonian simulation by quantum signal processing. *Physical Review Letters* **118**, 010501 (2017).

[37] Low, G. H. & Chuang, I. L. Hamiltonian simulation by qubitization. *Quantum* **3**, 163 (2019).

[38] Campbell, E. T. Random compiler for fast Hamiltonian simulation. *Physical Review Letters* **123**, 070503 (2019).

[39] Ouyang, Y., White, D. R. & Campbell, E. T. Compilation by stochastic Hamiltonian sparsification. *Quantum* **4**, 235 (2020).

[40] Childs, A. M., Ostrander, A. & Su, Y. Faster quantum simulation by randomization. *Quantum* **3**, 182 (2019).

[41] Chen, C.-F., Huang, H.-Y., Kueng, R. & Tropp, J. A. Concentration for random product formulas. *PRX Quantum* **2**, 040305 (2021).

[42] Magnus, W. On the exponential solution of differential equations for a linear operator. *Communications on Pure and Applied Mathematics* **7**, 649–673 (1954).

[43] Blanes, S., Casas, F., Oteo, J. A. & Ros, J. The Magnus expansion and some of its applications. *Physics Reports* **470**, 151–238 (2009).

[44] Sharma, K. & Tran, M. C. Hamiltonian simulation in the interaction picture using the Magnus expansion. *Physical Review Research* **6**, 033047 (2024).

[45] Fang, D., Liu, D. & Sarkar, R. Time-dependent hamiltonian simulation via Magnus expansion: Algorithm and superconvergence. *Communications in Mathematical Physics* **406**, 128 (2025).

[46] Cirstoiu, C. *et al.* Variational fast forwarding for quantum simulation beyond the coherence time. *npj Quantum Information* **6**, 82 (2020).

[47] Kökçü, E. *et al.* Fixed depth hamiltonian simulation via Cartan decomposition. *Physical Review Letters* **129**, 070501 (2022).

[48] Steckmann, T. *et al.* Simulating the Mott transition on a noisy digital quantum computer via Cartan-based fast-forwarding circuits. *PRX Quantum* **4**, 010308 (2023).

[49] Wu, S.-X., Du, S.-W., Fang, D. & Liu, D.-S. Randomized hamiltonian simulation by magnetic groups. *Physical Review A* **108**, 022418 (2023).

[50] Zeier, R. & Schulte-Herbrüggen, T. Symmetry principles in quantum systems theory. *Journal of Mathematical Physics* **52**, 113510 (2011).

[51] Sachdev, S. *Quantum Phase Transitions* (Cambridge University Press, Cambridge, 2011), 2nd edn.

[52] Cervera-Lierta, A. Exact Ising model simulation on a quantum computer. *Quantum* **2**, 114 (2018).

[53] Bravyi, S. B. & Kitaev, A. Y. Fermionic quantum computation. *Annals of Physics* **298**, 210–226 (2002).

[54] Whitfield, J. D., Biamonte, J. & Aspuru-Guzik, A. Simulation of electronic structure Hamiltonians using quantum computers. *Molecular Physics* **109**, 735–750 (2011).

[55] Seeley, J. T., Richard, M. J. & Love, P. J. The Bravyi-Kitaev transformation for quantum computation of electronic structure. *Journal of Chemical Physics* **137**, 224109 (2012).

[56] Verstraete, F. & Cirac, J. I. Mapping local Hamiltonians of fermions to local Hamiltonians of spins. *Journal of Statistical Mechanics: Theory and Experiment* **2005**, P09012 (2005).

[57] Cade, C., Mineh, L., Montanaro, A. & Sherborn, S. Strategies for solving the Fermi-Hubbard model on near-term quantum computers. *Physical Review B* **102**, 235122 (2020).

[58] Jordan, S. P., Lee, K. S. M. & Preskill, J. Quantum algorithms for quantum field theories. *Science* **336**, 1130–1133 (2012).

[59] Babbush, R. *et al.* Encoding electronic spectra in quantum circuits with linear T complexity. *Physical Review X* **8**, 041015 (2018).

[60] Babbush, R., Berry, D. W., McClean, J. R. & Neven, H. Quantum simulation of chemistry with sublinear scaling in basis size. *npj Quantum Information* **5**, 92 (2019).

[61] Peruzzo, A. *et al.* A variational eigenvalue solver on a photonic chip. *Nature Communications* **5**, 4213 (2014).

[62] McClean, J. R., Romero, J., Babbush, R. & Aspuru-Guzik, A. The theory of variational hybrid quantum-classical algorithms. *New Journal of Physics* **18**, 023023 (2016).

[63] Farhi, E., Goldstone, J. & Gutmann, S. A quantum approximate optimization algorithm (2014). 1411.4028.

[64] Cerezo, M. *et al.* Variational quantum algorithms. *Nature Reviews Physics* **3**, 625–644 (2021).

[65] Bharti, K. *et al.* Noisy intermediate-scale quantum algorithms. *Reviews of Modern Physics* **94**, 015004 (2022).

[66] Endo, S., Cai, Z., Benjamin, S. C. & Yuan, X. Hybrid quantum-classical algorithms and quantum error mitigation. *Journal of the Physical Society of Japan* **90**, 032001 (2021).

[67] Preskill, J. Quantum computing in the NISQ era and beyond. *Quantum* **2**, 79 (2018).

[68] Kandala, A. *et al.* Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature* **549**, 242–246 (2017).

[69] Kim, Y. *et al.* Evidence for the utility of quantum computing before fault tolerance. *Nature* **618**, 500–505 (2023).

3 Methods

3.1 Trotter–Suzuki steps and residual construction

For $k \geq 2$, higher even orders are generated by the Suzuki recursion

$$S_{2k}(\delta t) = S_{2k-2}(p_k \delta t)^2 S_{2k-2}((1-4p_k)\delta t) S_{2k-2}(p_k \delta t)^2, \qquad p_k = \left(4 - 4^{1/(2k-1)}\right)^{-1}. \tag{4}$$

For a fixed order q and step size δt, the residual factor is $R_q(\delta t) = U(\delta t) S_q(\delta t)^\dagger$, the corrected one-step propagator is $G_q(\delta t) = R_q(\delta t) S_q(\delta t)$, and the global approximation is $\widetilde{U}_{R,q}(t;r) = G_q(t/r)^r$. When R_q is computed exactly, $G_q(\delta t) = U(\delta t)$. Because R_q is unitary, it is represented as $R_q(\delta t) = e^{-iK_q(\delta t)}$ for a Hermitian residual generator K_q. The generator is not unique globally because eigenphases are defined modulo 2π; for small enough δt the residual lies close to the identity and the principal generator is well defined.

A compressed residual generator is $\widehat{K}_q(\delta t) \in \mathcal{B}$, where $\mathcal{B} \subseteq \mathsf{Herm}(d)$ is a real linear space of implementable Hermitian generators (Pauli strings of weight at most w, geometrically local Pauli strings of bounded diameter, commutator monomials up to a fixed degree, or generators representable by a fixed hardware ansatz), with $\widehat{R}_q(\delta t) = e^{-i\widehat{K}_q(\delta t)}$ and $\widehat{G}_q(\delta t) = \widehat{R}_q(\delta t) S_q(\delta t)$. The most direct dense benchmark instantiation is Pauli projection. With $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$ and $\mathcal{B}_w = \mathrm{span}_{\mathbb{R}}\{P \in \mathcal{P}_n : \mathrm{wt}(P) \leq w\}$, the projection is

$$\Pi_w K_q = \sum_{\mathrm{wt}(P) \leq w} \frac{\mathrm{Tr}(P K_q)}{2^n} P. \tag{5}$$

3.2 Theory: proofs

This section contains the main mathematical guarantees. All norms are spectral norms unless explicitly stated otherwise.

Lemma 1 (Unitarity of product-formula steps)**.** *Let A and B be Hermitian and let every Suzuki coefficient be real. Then $S_q(\delta t)$ is unitary for every real δt and every $q \in \{1, 2, 4, 6, 8\}$ generated by the Strang step and Eq. (4).*

Proof. Each elementary factor has the form $e^{-iC\tau}$ with $C = C^\dagger$ and $\tau \in \mathbb{R}$. Such a factor is unitary because $(e^{-iC\tau})^\dagger = e^{iC\tau} = (e^{-iC\tau})^{-1}$. The product of finitely many unitary matrices is unitary. The recursive Suzuki construction only composes such factors with real rescaled times, including the negative coefficient $1 - 4p_k$, which still gives a unitary exponential. □

Theorem 1 (Exact residual factorization and uniqueness)**.** *Let $H = H^\dagger$ and let $S_q(\delta t)$ be any unitary product-formula step. The residual $R_q(\delta t) = U(\delta t)S_q(\delta t)^\dagger$ is unitary and is the unique matrix L satisfying $LS_q(\delta t) = U(\delta t)$. Consequently, $R_q(\delta t)S_q(\delta t) = U(\delta t)$ and $[R_q(t/r)S_q(t/r)]^r = U(t)$ for time-independent H.*

Proof. Since H is Hermitian, $U(\delta t)$ is unitary. Lemma 1 gives unitarity of S_q, hence $S_q^\dagger = S_q^{-1}$. Thus $R_q^\dagger R_q = S_q U^\dagger U S_q^\dagger = S_q S_q^\dagger = \mathbb{I}$. Moreover, $R_q S_q = U S_q^\dagger S_q = U$. If $LS_q = U$, right multiplication by $S_q^\dagger$ gives $L = US_q^\dagger = R_q$. For $t = r\delta t$, time independence gives $U(\delta t)^r = U(r\delta t) = U(t)$. □

Theorem 2 (Optimality among all left corrections)**.** *Let $|||\cdot|||$ be any unitarily invariant norm. The residual R_q is a global minimizer of $\mathcal{J}(L) = |||U(\delta t) - LS_q(\delta t)|||$. The minimum value is zero, and the minimizer is unique.*

Proof. Every norm is nonnegative, so $\mathcal{J}(L) \geq 0$. Theorem 1 gives $\mathcal{J}(R_q) = 0$. If another L' also gives zero, then $L'S_q = U$, and uniqueness in Theorem 1 implies $L' = R_q$. □

Remark 1. *Theorem 2 is an oracle statement, not a cost statement. It says that no exact left multiplier can beat R_q in error because the error is already zero. It does not say that R_q is cheap to implement.*

Theorem 3 (Small-step residual-generator bound)**.** *Assume S_q is order q in the sense that there exist constants C_q and δt_0 such that $\|U(\delta t) - S_q(\delta t)\|_2 \leq C_q |\delta t|^{q+1}$ for $|\delta t| \leq \delta t_0$. If $C_q |\delta t|^{q+1} \leq 1$, then $R_q(\delta t)$ has a Hermitian logarithm $K_q(\delta t)$ with phases chosen in $[-\pi, \pi]$ and $\|K_q(\delta t)\|_2 \leq 2C_q |\delta t|^{q+1}$.*

Proof. Because S_q is unitary, $\|R_q - \mathbb{I}\|_2 = \left\|US_q^\dagger - \mathbb{I}\right\|_2 = \|U - S_q\|_2 \leq C_q |\delta t|^{q+1}$. Diagonalize $R_q = W \operatorname{diag}(e^{-i\theta_j})W^\dagger$ with $\theta_j \in [-\pi, \pi]$ and define $K_q = W \operatorname{diag}(\theta_j)W^\dagger$. For any eigenphase, $\left|e^{-i\theta_j} - 1\right| = 2|\sin(\theta_j/2)|$. If $\|R_q - \mathbb{I}\|_2 \leq 1$, then $|\theta_j| \leq 2\arcsin(\|R_q - \mathbb{I}\|_2/2) \leq 2\|R_q - \mathbb{I}\|_2$. Taking the maximum over j proves the bound. □

Theorem 4 (Multi-step stability for approximate residuals)**.** *Let $\widehat{R}_q$ be any approximate residual and define $\widehat{G}_q = \widehat{R}_q S_q$. If $\eta = \left\|\widehat{R}_q - R_q\right\|_2$, then*

$$\left\|\widehat{G}_q^{\,r} - U(t)\right\|_2 \leq \eta \sum_{j=0}^{r-1} (1+\eta)^j \leq r\eta e^{(r-1)\eta}. \tag{6}$$

If $\widehat{R}_q$ is unitary, then the sharper bound $\left\|\widehat{G}_q^{\,r} - U(t)\right\|_2 \leq r\eta$ holds.

Proof. Since S_q is unitary, $\left\|\widehat{G}_q - G_q\right\|_2 = \left\|(\widehat{R}_q - R_q)S_q\right\|_2 = \eta$. Also $G_q = U(\delta t)$ and $\|G_q\|_2 = 1$. The telescoping identity $\widehat{G}_q^{\,r} - G_q^r = \sum_{j=0}^{r-1} \widehat{G}_q^{\,r-1-j}(\widehat{G}_q - G_q)G_q^j$ combined with submultiplicativity gives Eq. (6), because $\left\|\widehat{G}_q\right\|_2 \le \|G_q\|_2 + \eta = 1 + \eta$. If $\widehat{R}_q$ is unitary, then $\widehat{G}_q$ is unitary and every factor in the telescoping sum has norm one except $\widehat{G}_q - G_q$, giving the sharper bound. □

Theorem 5 (Tightness of the multi-step stability bound)**.** *Let $\delta t > 0$, let $H = H^\dagger$, and let $S_q(\delta t)$ be an order-q step with exact residual R_q and corrected step $G_q = R_q S_q = U(\delta t)$. For every $\eta \in (0, \pi]$ there exists a* unitary *approximate residual $\widehat{R}_q$ with $\left\|\widehat{R}_q - R_q\right\|_2 \le \eta$ such that, with $\widehat{G}_q = \widehat{R}_q S_q$,*

$$\left\|\widehat{G}_q^{\,r} - U(r\delta t)\right\|_2 \ge \tfrac{2}{\pi}\, r\eta \qquad \textit{for every integer } r \textit{ with } r\eta \le \pi. \tag{7}$$

Hence the linear-in-r estimate of Theorem 4 is order-optimal: no bound of the form $o(r\eta)$ can hold uniformly in r.

Proof. Let $P = P^\dagger$ be a spectral projector of H onto a single eigenspace, so $\|P\|_2 = 1$ and $[P, U(s)] = 0$ for all real s. Set $\widehat{R}_q = e^{-i\eta P}R_q$, which is unitary. Since R_q is unitary, $\left\|\widehat{R}_q - R_q\right\|_2 = \left\|e^{-i\eta P} - \mathbb{I}\right\|_2 = \max_{p\in\{0,1\}} \left|e^{-i\eta p} - 1\right| = 2\sin(\eta/2) \le \eta$. Because $R_q S_q = U(\delta t)$ and $[P, U(\delta t)] = 0$,

$$\widehat{G}_q^{\,r} = \left(e^{-i\eta P}U(\delta t)\right)^r = e^{-ir\eta P}U(\delta t)^r = e^{-ir\eta P}U(r\delta t).$$

Therefore $\left\|\widehat{G}_q^{\,r} - U(r\delta t)\right\|_2 = \left\|e^{-ir\eta P} - \mathbb{I}\right\|_2 = 2\sin(r\eta/2)$. For $r\eta \le \pi$, Jordan's inequality $\sin x \ge \frac{2}{\pi}x$ on $[0, \pi/2]$ gives $2\sin(r\eta/2) \ge \frac{2}{\pi} r\eta$. □

Theorem 6 (Duhamel bound for generator approximation)**.** *Let K and $\widehat{K}$ be Hermitian. Then $\left\|e^{-iK} - e^{-i\widehat{K}}\right\|_2 \le \left\|K - \widehat{K}\right\|_2$. Consequently, if $R_q = e^{-iK_q}$ and $\widehat{R}_q = e^{-i\widehat{K}_q}$, then $\left\|\widehat{G}_q^{\,r} - U(t)\right\|_2 \le r\left\|K_q - \widehat{K}_q\right\|_2$.*

Proof. Define $F(s) = e^{-i[(1-s)K + s\widehat{K}]}$. The derivative is represented by the standard Duhamel formula $\frac{dF}{ds} = -i\int_0^1 e^{-i(1-a)H_s}(\widehat{K} - K)e^{-iaH_s}\, da$, where $H_s = (1-s)K + s\widehat{K}$. Since H_s is Hermitian, the exponentials in the integral are unitary, so $\|dF/ds\|_2 \le \left\|\widehat{K} - K\right\|_2$. Integrating from $s = 0$ to $s = 1$ proves the first bound. The global bound follows from Theorem 4, since both exponentials are unitary. □

Theorem 7 (Frobenius-optimal Pauli compression)**.** *Let $K = K^\dagger$ on n qubits and let $\mathcal{B} \subseteq \mathrm{span}_{\mathbb{R}}(\mathcal{P}_n)$ be any real Pauli subspace. The orthogonal projection $\Pi_{\mathcal{B}}K = \sum_{P\in\mathcal{B}} \frac{\mathrm{Tr}(PK)}{2^n}P$ uniquely minimizes $\|K - L\|_F$ over $L \in \mathcal{B}$. The corresponding unitary residual approximation satisfies $\left\|e^{-iK} - e^{-i\Pi_{\mathcal{B}}K}\right\|_2 \le \|K - \Pi_{\mathcal{B}}K\|_2 \le \|K - \Pi_{\mathcal{B}}K\|_F$.*

Proof. The normalized Pauli operators $P/\sqrt{2^n}$ form an orthonormal basis under the Hilbert–Schmidt inner product. Orthogonal projection onto a closed finite-dimensional subspace is the unique Frobenius-norm minimizer. The spectral norm is bounded by the Frobenius norm, and Theorem 6 supplies the unitary perturbation bound. □

Corollary 1 (Projected residual simulation certificate)**.** *For $K_{q,w} = \Pi_w K_q$ and $\widehat{G}_{q,w} = e^{-iK_{q,w}}S_q$, $\left\|\widehat{G}_{q,w}^{\,r} - U(t)\right\|_2 \le r\,\|K_q - K_{q,w}\|_2$. Thus a target global accuracy ϵ is guaranteed whenever $\|K_q - K_{q,w}\|_2 \le \epsilon/r$.*

Theorem 8 (Leading Strang residual support for the open-boundary TFIM)**.** *Let A and B be the TFIM splitting in Eq. (3). For the Strang step $S_2(\delta t)$, the degree-three homogeneous term in the residual generator $K_2(\delta t)$ is a real linear combination of Pauli strings with weight at most three. Equivalently,*

$$K_2(\delta t) = \delta t^3 K_2^{(3)} + O(\delta t^5), \qquad K_2^{(3)} \in \mathcal{B}_3. \tag{8}$$

Proof. The symmetric BCH expansion of Strang splitting has no degree-two term; the first nonzero defect is a linear combination of double commutators $[A,[A,B]]$ and $[B,[A,B]]$ [42, 43]. The residual generator is the negative of this leading logarithmic defect up to the same order, so it is enough to bound the Pauli weight of these double commutators.

Write $A = \sum_e A_e$ with $A_e = JZ_jZ_{j+1}$ for edges $e = (j, j+1)$ and $B = \sum_v B_v$ with $B_v = hX_v$. A commutator $[A_e, B_v]$ vanishes unless v is incident to e. If it is nonzero, its support is contained in the two sites of e and has Pauli weight two. In $[B,[A,B]]$, the second commutator adds at most one on-site term $B_{v'}$ whose support must overlap the existing support; therefore the support remains within the same edge and the Pauli weight is at most two. In $[A,[A,B]]$, the second edge term $A_{e'}$ must overlap the support of $[A_e, B_v]$. The union of two overlapping nearest-neighbor edges in a one-dimensional open chain contains at most three sites. Therefore every nonzero Pauli string in the double commutators has weight at most three. This proves $K_2^{(3)} \in \mathcal{B}_3$. □

Theorem 9 (Size-transfer error bound for learned local residual generators)**.** *Consider the disordered open-boundary TFIM on n sites with couplings in a fixed compact set, and let $K_2^{(3)}(n) = \sum_{j=1}^n h_j$ be the degree-three Strang residual generator, where by Theorem 8 each h_j is a real linear combination of weight-≤ 3 Pauli strings supported in the three-site window $\{j-1, j, j+1\}$ (truncated at the boundaries). Suppose a translation-invariant per-site predictor outputs $\widehat{h}_j$ from the couplings in that window with $\left\|\widehat{h}_j - h_j\right\|_2 \leq \eta$ for every j, where η depends only on the predictor and the coupling range, not on n. Then $\widehat{K}_2^{(3)}(n) = \sum_j \widehat{h}_j$ satisfies $\left\|\widehat{K}_2^{(3)}(n) - K_2^{(3)}(n)\right\|_2 \leq n\,\eta$, and the corrected step $\widehat{G}(\delta t) = \exp(-i\delta t^3 \widehat{K}_2^{(3)})\, S_2(\delta t)$ obeys, for $t = r\delta t$, $\left\|\widehat{G}(\delta t)^r - U(t)\right\|_2 \leq r\,\delta t^3 n\,\eta + r\,O(\delta t^5)$.*

Proof. The first bound is the triangle inequality over the n site-anchored terms, $\left\|\sum_j(\widehat{h}_j - h_j)\right\|_2 \leq \sum_j \left\|\widehat{h}_j - h_j\right\|_2 \leq n\eta$; translation invariance makes the per-site bound η independent of n. For the second, set $\widehat{K} = \delta t^3 \widehat{K}_2^{(3)}$; the BCH expansion of Appendix 3.3 gives $K_2(\delta t) = \delta t^3 K_2^{(3)} + O(\delta t^5)$, so $\left\|\widehat{K} - K_2(\delta t)\right\|_2 \leq \delta t^3 n\eta + O(\delta t^5)$. The Duhamel bound (Theorem 6) gives $\left\|e^{-i\widehat{K}} - R_2\right\|_2 \leq \left\|\widehat{K} - K_2\right\|_2$, and the multi-step stability bound (Theorem 4) multiplies by r, giving the estimate. □

Theorem 10 (From per-site coefficient error to a global certificate)**.** *Adopt the setting of Theorem 9 and expand each per-site generator in the fixed family of T weight-≤ 3 Pauli templates anchored at site j, $h_j = \sum_{\alpha=1}^T c_{j,\alpha} P_{j,\alpha}$ and $\widehat{h}_j = \sum_{\alpha=1}^T \widehat{c}_{j,\alpha} P_{j,\alpha}$, where each $P_{j,\alpha}$ is a Pauli string, so $\|P_{j,\alpha}\|_2 = 1$. If the per-site coefficient error obeys $\|c_j - \widehat{c}_j\|_2 \leq \delta_c$ for every j, then the per-site operator error of Theorem 9 satisfies $\eta \leq \sqrt{T}\,\delta_c$, and the learned corrected step $\widehat{G}(\delta t) = \exp(-i\delta t^3 \widehat{K}_2^{(3)})\, S_2(\delta t)$ obeys, for $t = r\delta t$,*

$$\left\|\widehat{G}(\delta t)^r - U(t)\right\|_2 \leq r\,\delta t^3\, n\sqrt{T}\,\delta_c + r\,O(\delta t^5). \tag{9}$$

Proof. By the triangle inequality and $\|P_{j,\alpha}\|_2 = 1$,

$$\left\|\widehat{h}_j - h_j\right\|_2 = \left\|\sum_{\alpha=1}^{T}(\widehat{c}_{j,\alpha} - c_{j,\alpha})P_{j,\alpha}\right\|_2 \le \sum_{\alpha=1}^{T}|\widehat{c}_{j,\alpha} - c_{j,\alpha}| = \|c_j - \widehat{c}_j\|_1 \le \sqrt{T}\,\|c_j - \widehat{c}_j\|_2 \le \sqrt{T}\,\delta_c,$$

where the last line uses the Cauchy–Schwarz inequality $\|x\|_1 \le \sqrt{T}\,\|x\|_2$ on $\mathbb{R}^T$. Taking the maximum over j gives $\eta \le \sqrt{T}\,\delta_c$; substituting this into the size-transfer bound of Theorem 9 yields Eq. (9). □

Remark 2. *With the $T = 48$ contiguous weight-≤ 3 templates per site used in Sec. 1.5 and the measured held-out coefficient accuracy (relative ℓ_2 error 1.43×10^{-2}, $R^2 = 0.9998$), Theorem 10 converts the learned-coefficient accuracy directly into the* a priori *global error bound whose linear-in-r growth Fig. 3c verifies empirically.*

Remark 3. *Theorem 8 does not say the full residual generator lies in $\mathcal{B}_3$. Higher-order BCH terms can produce larger support. It predicts that $w \le 3$ projection should remove the dominant Strang defect at small step size, while $w \le 4$ and higher should capture additional subleading structure. This is consistent with the numerical results reported above.*

Theorem 11 (A posteriori step certificate)**.** *Let $\widehat{G}$ be any unitary approximate one-step propagator and let $\Delta = \left\|\widehat{G} - U(\delta t)\right\|_2$. Then $\left\|\widehat{G}^r - U(t)\right\|_2 \le r\Delta$.*

Proof. Apply the telescoping identity to $\widehat{G}^r - U(\delta t)^r$. Both $\widehat{G}$ and $U(\delta t)$ are unitary, so each of the r terms is bounded by Δ. □

3.3 Numerical methodology

All numerical results are generated by deterministic dense-matrix scripts that build the Pauli matrices, construct A, B, and H, evaluate matrix exponentials using dense linear algebra, compute spectral norms from singular values, and write the resulting CSV, JSON, PDF, and table files. There are no hand-coded performance numbers in the plotting pipeline. The deterministic studies use fixed grids with no stochastic sampling. The learned-residual experiment (Sec. 1.5) does draw disordered couplings and train a network, but it is made reproducible by fixed random seeds for the sampler, the data split, and the network initialization; its training labels and all reported error metrics are computed by the same dense-matrix code, so rerunning the script reproduces every reported value. The benchmark design deliberately includes checks that would expose fabrication or implementation errors: unitarity checks for S_q, R_q, and corrected steps; same-order comparisons between S_q and R_qS_q; exact dense recomputation of $U(t)$ rather than relying on repeated one-step products as the reference; saved raw data for every plotted point; and finite-precision reporting, including cases where round-off makes the high-order baseline comparable to the oracle residual.

The Hilbert-space dimension is $d = 2^n$. For the fixed-time study we use $J = h = 1$, $t = 1$, and $r = 10$ steps. The exact total propagator $U(t)$ is computed as $\exp(-iHt)$ independently of the product-formula steps. For the projected residual study we compute the residual generator $K_q = i\log(R_q)$ using the dense matrix logarithm, symmetrize K_q as $(K_q + K_q^\dagger)/2$ to remove round-off skew-Hermitian contamination, and project onto Pauli strings of weight at most w using Eq. (5). The projected corrected propagator is $[e^{-i\Pi_w K_q}S_q]^r$. The learned-residual study additionally trains small multilayer perceptrons (on global-oracle and on oracle-free local-patch labels) with a standard automatic-differentiation library on disordered chains; all couplings, step sizes, and network weights are seeded from a single fixed seed, and every learned generator is scored only through the dense spectral-norm metrics above. Each network is trained for 4000 full-batch Adam steps on 220 disordered realizations at each of $n = 4$ and $n = 5$; reported transfer errors are averaged over independent held-out realizations at each size (40 realizations for $n = 4$–7, 24 for $n = 8$, 16 for $n = 9$, 12 for $n = 10$), and the shaded bands in Fig. 3 report one standard deviation across these realizations. All plotted errors are spectral-norm errors. Where a plotted oracle residual appears around 10^{-14} to 10^{-13}, that value is the observed double-precision floor, not a mathematical error in exact arithmetic.

Data availability

All data supporting the numerical findings are generated by deterministic scripts from the Hamiltonian definition in Eq. (3). The raw CSV and JSON files, figure-generation scripts, and table-generation scripts are included with the manuscript package. No experimental result is fabricated or manually inserted without a corresponding generated data record.

Code availability

The accompanying code constructs Pauli operators, Trotter–Suzuki steps, exact propagators, residual generators, Pauli projections, tables, and figures from first principles. The code is intended as a reproducibility baseline and as a starting point for GPU-accelerated implementations.

Author contributions

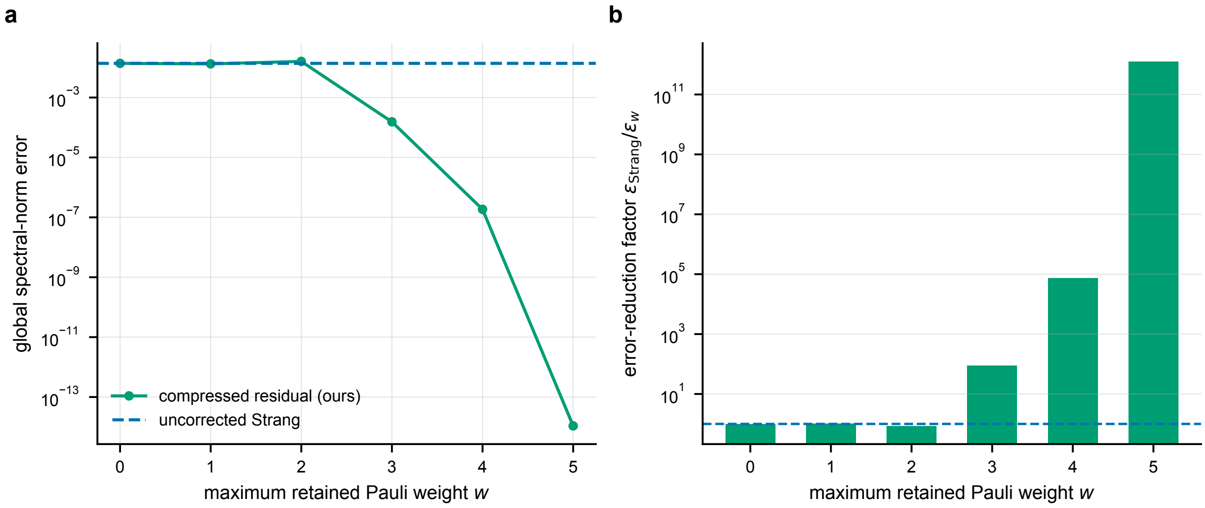
M.H. conceived the residual-generator framework, developed and proved the theoretical results, designed and implemented the dense-matrix and operator-learning experiments, produced all figures and tables, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Acknowledgements

The author acknowledges the foundational literature on product formulas, Hamiltonian simulation, commutator error bounds, quantum dynamics algorithms, and quantum-circuit compilation that motivates this work.



Extended Data Fig. 1 | Compressed-residual compilation for $n = 5$, $q = 2$ ($J = h = 1$, $t = 1$, $r = 10$). **a**, Spectral-norm error versus maximum retained Pauli weight w, against the uncorrected baseline. **b**, Improvement factor over the baseline by weight. The sharp improvement at $w = 3$ is predicted by the leading-support theorem for the Strang residual (Theorem 8). All values are exact, deterministic dense-matrix computations (no sampling, hence no error bars); the $w = 5$ row sits at the double-precision floor.

Table 3. Weight-truncated residual compilation for $n = 5$, $q = 2$, $J = h = 1$, $t = 1$, and $r = 10$. The column w keeps Pauli strings of weight at most w in the exact residual generator. Every entry is a single exact, deterministic dense-matrix computation (no statistical sampling, hence no uncertainty interval); the $w = 5$ row is at the double-precision floor.

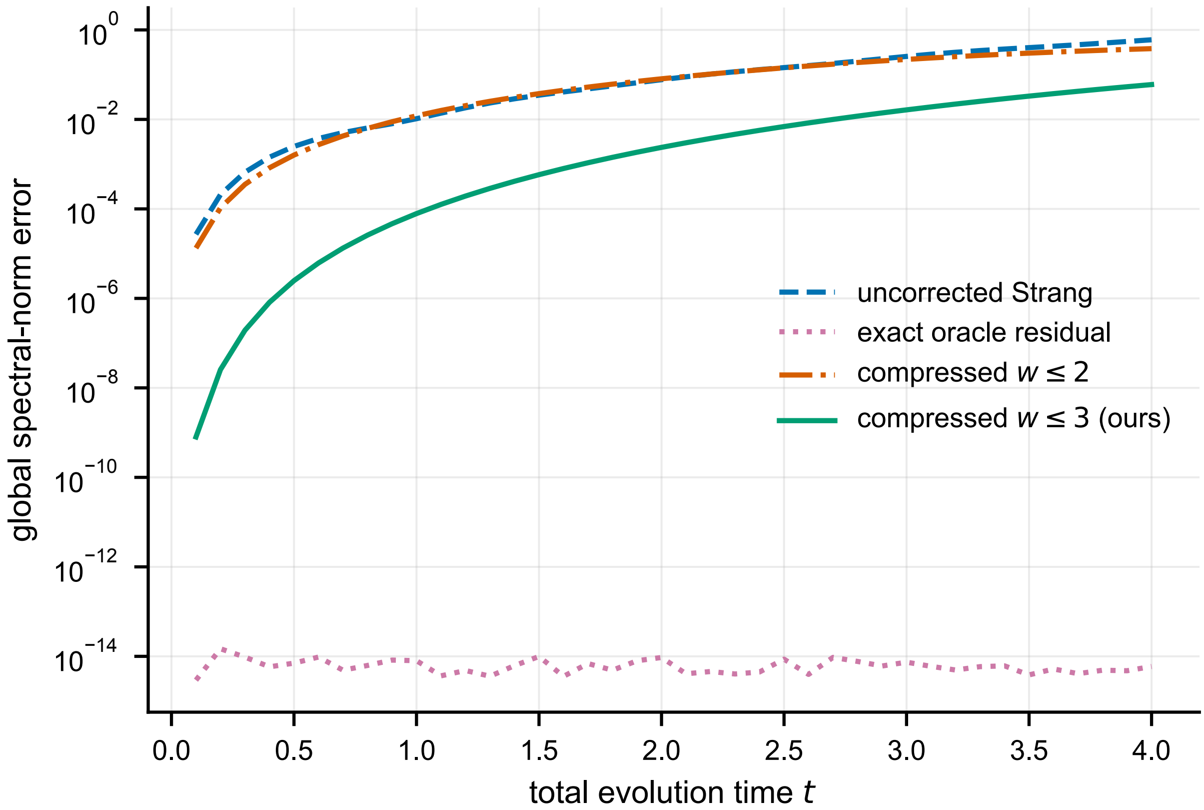
w	$\lVert K_q - \Pi_w K_q \rVert_2$	projected error	improvement
0	$3.656e-3$	$1.384e-2$	$1.00\times$
1	$2.859e-3$	$1.330e-2$	$1.04\times$
2	$1.949e-3$	$1.607e-2$	$0.86\times$
3	$1.573e-5$	$1.545e-4$	$89.56\times$
4	$1.850e-8$	$1.850e-7$	$74829.05\times$
5	$1.143e-18$	$1.089e-14$	$1270805509048.55\times$

Extended Data

Supplementary theory: detailed BCH support analysis for the TFIM

The support theorem in the main text is intentionally stated without relying on exact BCH coefficients. This appendix gives the more explicit structure. For Strang splitting, $S_2(\delta t) = e^{-iB\delta t/2}e^{-iA\delta t}e^{-iB\delta t/2}$, the logarithm has the form $\log S_2(\delta t) = -i(A+B)\delta t + \delta t^3\Gamma_3 + O(\delta t^5)$, where Γ_3 is a linear combination of $[A,[A,B]]$ and $[B,[A,B]]$. The absence of an even power follows from time-reversal symmetry of the Strang formula. The residual is $R_2(\delta t) = e^{-i(A+B)\delta t}S_2(\delta t)^\dagger$, and the leading residual generator cancels the leading logarithmic defect, so $K_2(\delta t) = \delta t^3 K_2^{(3)} + O(\delta t^5)$, where $K_2^{(3)}$ is a real linear combination of $i[A,[A,B]]$ and $i[B,[A,B]]$ after accounting for the anti-Hermitian nature of commutators of Hermitian matrices.

For the TFIM, write $A_e = JZ_jZ_{j+1}$ and $B_v = hX_v$. The elementary commutator is $[A_e, B_v] = 0$ if $v \notin e$. If $v = j$, then $[Z_jZ_{j+1}, X_j] = 2iY_jZ_{j+1}$, and if $v = j+1$, then $[Z_jZ_{j+1}, X_{j+1}] = 2iZ_jY_{j+1}$. Thus $[A,B]$ is a sum of weight-two Pauli strings on nearest-neighbor edges. Next consider $[B,[A,B]]$: a term $[B_{v'}, [A_e, B_v]]$ is nonzero only if v' lies in the support of

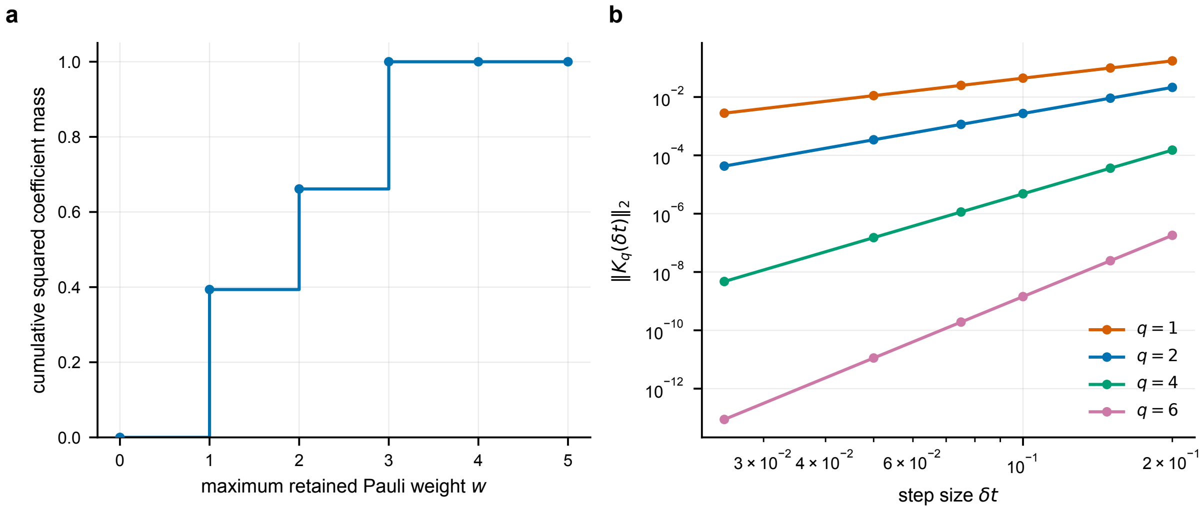


Extended Data Fig. 2 | Time-resolved correction hierarchy for $n = 4$, $q = 2$, $J = h = 1$, and $r = 10$, over 40 total times $t \in [0.1, 4.0]$. The $w \leq 3$ projected residual uniformly improves the baseline in this sweep; $w \leq 2$ is less reliable. Each curve is an exact, deterministic dense-matrix computation (no sampling, hence no error bars).

$[A_e, B_v]$, which is the same edge e; the resulting Pauli string remains supported on that edge and has weight at most two, for example $[X_j, Y_j Z_{j+1}] = 2iZ_j Z_{j+1}$ and $[X_{j+1}, Y_j Z_{j+1}] = -2iY_j Y_{j+1}$. Now consider $[A, [A, B]]$: a term $[A_{e'}, [A_e, B_v]]$ is nonzero only if e' overlaps the support of $[A_e, B_v]$. In a one-dimensional nearest-neighbor chain, the union of two overlapping edges contains at most three sites, $(j-1, j, j+1)$ or $(j, j+1, j+2)$. Therefore every nonzero Pauli string in $[A, [A, B]]$ has weight at most three, proving that the leading Strang residual generator is contained in $\mathcal{B}_3$. The same proof strategy extends to other local Hamiltonians: if the interaction hypergraph has local terms of diameter at most r_0, then an m-fold nested commutator can only grow along overlapping local supports, connecting residual-generator compression to locality bounds and Lieb–Robinson ideas [29, 30].

Supplementary theory: projection formulas and diagnostic quantities

For n qubits, the Pauli group modulo phases gives 4^n Hermitian strings satisfying $\mathrm{Tr}(PQ) = 2^n \delta_{P,Q}$. Every Hermitian matrix K has an expansion $K = \sum_{P \in \mathcal{P}_n} c_P P$ with $c_P = \mathrm{Tr}(PK)/2^n$, squared Frobenius norm $\|K\|_F^2 = 2^n \sum_{P \in \mathcal{P}_n} |c_P|^2$, and cumulative coefficient mass (plotted in Extended Data Fig. 3a) $M(w) = \sum_{\mathrm{wt}(P) \leq w} |c_P|^2 / \sum_{P \in \mathcal{P}_n} |c_P|^2$. For the $n = 5$, $q = 2$, $\delta t = 0.1$ run, the generated squared coefficient masses by weight are $M'_0 = 3.598 \times 10^{-33}$, $M'_1 = 1.466 \times 10^{-6}$, $M'_2 = 9.973 \times 10^{-7}$, $M'_3 = 1.262 \times 10^{-6}$, $M'_4 = 1.235 \times 10^{-10}$, and $M'_5 = 3.413 \times 10^{-16}$. Although the weight-four mass is small in Frobenius terms, it can still matter for spectral-norm accuracy after repeated steps, which is why Extended Data Table 3



Extended Data Fig. 3 | Structure of the residual generator. **a**, Cumulative squared Pauli coefficient mass for the $n = 5$, $q = 2$ residual generator at $\delta t = 0.1$; weight three captures the dominant leading structure, while higher weights capture subleading BCH terms. **b**, Residual-generator norm $\|K_q(\delta t)\|_2$ versus step size for $n = 4$ and $q = 1, 2, 4, 6$; the norm decreases rapidly with order, consistent with product-formula order conditions and Theorem 3. Both panels are exact, deterministic dense-matrix computations (no sampling, hence no error bars).

Table 4. Per-step resource proxy. The baseline factor count is the number of elementary $e^{-iA\tau}$ or $e^{-iB\tau}$ exponentials in S_q. The oracle residual construction uses one additional dense exponential for $U(\delta t)$ and one dense multiplication by $S_q^\dagger$; it is not a scalable gate-level implementation. All entries are exact integer counts, not measured quantities.

q	Suzuki factors	dense $U(\delta t)$ expm	dense residual multiply
1	2	1	1
2	3	1	1
4	15	1	1
6	75	1	1
8	375	1	1

shows a large improvement from $w = 3$ to $w = 4$.

Supplementary theory: finite-precision interpretation

The exact residual theorem is an algebraic identity. In floating-point arithmetic, the computed residual is $\widetilde{R}_q = \mathrm{fl}\{\widetilde{U}(\delta t)\widetilde{S}_q(\delta t)^\dagger\}$ and the computed corrected step is $\widetilde{G}_q = \mathrm{fl}\{\widetilde{R}_q\widetilde{S}_q\}$. Round-off enters through every matrix exponential and multiplication. This explains why the $q = 8$ baseline, already extremely accurate at $\delta t = 0.1$, can appear slightly more accurate than the oracle residual in Table 1: the theorem says the exact-arithmetic error is zero, while the table reports floating-point evaluations of different arithmetic circuits. For low and moderate order, the product-formula error dominates round-off and oracle RGTC gives many orders of magnitude improvement; for very high order at small step size, product-formula error can be below or comparable to double-precision round-off, so differences at 10^{-13} are not physically meaningful; projected residual experiments should be judged at error levels well above round-off, such as the $w = 3$ and $w = 4$ results in Extended Data Table 3.

Supplementary: reproducibility checklist

A reproducible implementation should verify $\left\| S_q^\dagger S_q - \mathbb{I} \right\|_2 < 10^{-12}$ for every tested q, n, and δt; verify $\left\| R_q^\dagger R_q - \mathbb{I} \right\|_2 < 10^{-12}$ for every oracle residual; verify $\|R_q S_q - U(\delta t)\|_2 < 10^{-12}$ for small and moderate orders; save raw CSV or JSON data before plotting; make plots only from saved raw data, not from hard-coded arrays; record package versions and BLAS backend information; use deterministic parameter grids or fixed random seeds; and include tests that fail if a citation key in `main.tex` is missing from `refs.bib`. These checks are especially important if the code is ported to GPU backends, where matrix-exponential implementations and floating-point accumulation order may differ from CPU reference values.

Supplementary: possible extensions

The residual-generator framework suggests several concrete research extensions. A *BCH residual compiler* derives explicit nested-commutator formulas for K_q through a target order and implements $e^{-iK_q^{\mathrm{BCH}}}$ as a short product formula; Theorems 4 and 6 convert truncation error into global simulation error. A *variational residual compiler* uses the oracle residual on small systems as a target for a parameterized local circuit $V(\theta)$, with loss $\|V(\theta) - R_q\|_F^2$, local Pauli-shadow loss, or state-averaged infidelity, and the stability theorem provides the operator-norm target. An *operator-learning residual compiler* is realized in Sec. 1.5 for the Strang TFIM, learning the step-size-conditioned local weight-≤ 3 coefficients of K_2 with a single size-transferable network trained from oracle-free local patches; natural continuations extend the local templates to other lattices and higher orders, replace patch labels with classical-shadow or commutator estimates on hardware, and compile the learned generator $e^{-i\widehat{K}_q}$ into native gates using Theorems 4 and 6 as the operator-norm target. A *GPU-accelerated benchmark generator* uses CPU dense simulation as a correctness reference, then ports parameter sweeps and Pauli projections to batched GPU kernels, keeping output tolerance-comparable to the CPU reference and always regenerating raw data before producing figures.